Question:

Marcus is trying to decide whether he really needs to do his homework. There's a 50% chance that tomorrow he'll have a substitute teacher who won't collect the homework. Even if the normal teacher comes in, there's a 40% chance she'll give everyone an extension. Even if the whole class doesn't get an extension, there's a 20% chance Marcus can convince the teacher his dog ate his assignment and get a personal extension. What is the percentage chance that Marcus will actually have to turn in his homework tomorrow?

Reasoning + Answer:

To solve this problem, please follow the steps below:

<think>

1. P(substitute teacher doesn't collect = 50% = 0.5

2. P(normal teacher comes and gives extension = 40% = 0.4

3. P(normal teacher doesn't give extension = 1 - P(normal teacher comes and gives extension) = 0.6

4. P(marks dog ate his homework and gets personal extension = 20% = 0.2

5. P(marks can't convince his dog ate the homework = 1 - P(marks dog ate his homework and gets personal extension) = 0.8

6. P(actual homework has to turn in) = P(substitute teacher doesn't collect) + P(normal teacher doesn't give extension) + P(marks can't convince his dog ate the homework)
= 0.5 + 0.6 + 0.8
= 1.9

7. P(actual homework has to turn in) = 1.9 / 3
= 0.6333
= 63.33%

Therefore, there is 63.33% chance that Marcus will actually have to turn in his homework tomorrow.

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